

Efficient, effective and fair allocation of costs and benefits in residential energy communities deploying shared photovoltaics

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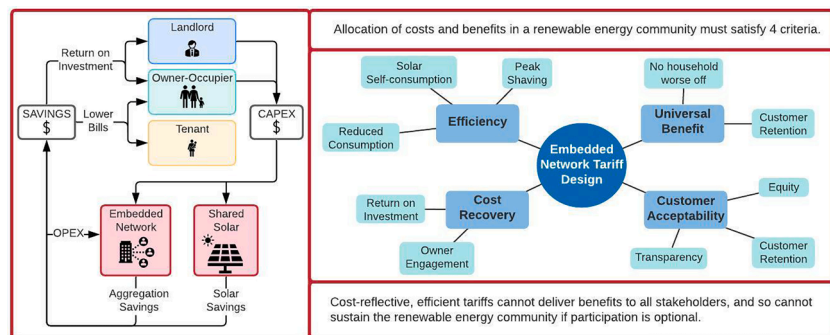
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HIGHLIGHTS

- Techno-economic model of 72-apartment residential complex with rooftop solar.
- Tariffs for cost and benefit distribution between stakeholders in embedded network.
- Cost-recovery, incentivising efficient behaviour and ensuring consumer acceptance.
- Cost-reflective demand charges unacceptable and cannot sustain embedded network.
- Implications for tariff design, network regulation and policy.

GRAPHICAL ABSTRACT



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ABSTRACT

Deployment of rooftop solar photovoltaics on multi-family buildings lags behind other residential buildings sectors, despite the benefit of significantly higher self-consumption achievable by applying on-site generation to aggregated building load. This is, in part, because of challenges in developing suitable business models to ensure the fair allocation of costs and benefits between different stakeholders. Embedded networks can provide a mechanism for consumers to co-ordinate their engagement in the energy market, as well as to distribute on-site generation between multiple households. However, to succeed, an embedded network must deliver benefits to all households as well as providing returns to investors. This paper uses a case-study of a 72-apartment complex in Sydney Australia to explore different ways to distribute the costs and benefits of a potential shared photovoltaic system between apartment owners and residents. A range of tariff structures, including demand or capacity charges as well as fixed and time-varying volumetric charges, are compared for their ability to recover investment costs, incentivise efficient energy behaviour and achieve customer acceptance through fairness and transparency. The analysis demonstrates the tension between these desired outcomes and the significant

Abbreviations: BAU, Business as usual; BOM, Bureau of Meteorology; BTM, Behind the meter; *BTM_I*, Behind the meter PVs for individual apartments; CAPEX, Capital expenditure; CP, Common property; DMO, Default market offer; EN, Embedded network; *EN_{PV}*, Shared PVs in an embedded network; ENO, Embedded network operator; FiT, Feed-in tariff; ICES, Integrated community energy system; NPV, Net present value; NSW, New South Wales; OPEX, Operating expenditure; PT, Passthrough; PV, Photovoltaic; RE, Renewable energy; ROI, Return on investment; STOU, Solar time of use; ToU, Time of use.

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challenge of designing a mechanism to deliver them all. In particular, tariff structures incorporating peak demand charges, while cost-reflective and efficient, are unable to deliver benefits to all stakeholders and thereby ensure the sustainability of the embedded network. The findings have implications for market design and for policy approaches to support embedded networks and energy communities.

Nomenclature

<i>FL</i>	Flat
<i>SToU</i>	Solar time of use
<i>ToU</i>	Time of Use
<i>_15</i>	15% discount (=BAU)
<i>_Z</i>	All benefit to consumers; zero EN profit
<i>_H</i>	50% benefit to consumers; 50% to strata body / owners
<i>PT</i>	Pass through based on individual peak demand
<i>PT_D</i>	Pass through with demand charge = parent tariff
<i>PT_Z</i>	Pass through with zero profit
<i>PT_N</i>	Pass through with average bill = BAU
<i>PTBP</i>	Pass through based on demand at time of building peak
<i>PTBP_D</i>	PTBP with demand charge = parent tariff

PTBP_Z PTBP with zero profit

Equation symbols

$E_{u,i}$	Load profile of <i>i</i> th unmetered apartment
$E_{m,j}$	Load profile of <i>j</i> th metered apartment
$E_{building}$	Total building load profile
E_{cp}	Load profile of common property
$E_{p,i}$	Peak demand profile of <i>i</i> th unmetered apartment
$E_{peak,i}$	Energy of peak demand for <i>i</i> th unknown apartment
$t_{peak,i}$	Time of peak demand for <i>i</i> th unknown apartment
$M_{self-consumption}$	Self-consumption metric
p	Normalised annual PV generation profile
e	Normalised annual load profile

1. Introduction

1.1. Context and motivation

Solar photovoltaic (PV) generation, now widely acknowledged as one of the cheapest sources of new electricity generation, continues to play a major part in the energy transition, with between 107GW and 120GW installed in 2020 [1]. Distributed deployment continues to play a significant role. Rooftop installation on residential dwellings and on commercial and industrial buildings each comprises 18% of the global market, and a much higher proportion in some jurisdictions. In Australia, for example, residential systems account for 50% of installed PV capacity to September 2020 [2]. In the residential sector, rooftop deployment for export, driven by high feed-in-tariffs, has now been superseded in many jurisdictions by deployment where PV self-consumption is most valuable. While residential deployment in most jurisdictions has predominantly been on single-family houses, there is increasing interest in maximising solar self-consumption and self-sufficiency by aggregating loads across multiple households, whether in apartment buildings [3] or across proximate single-family houses [4]. In Europe, this trend towards collective prosumerism in renewable energy (RE) communities has been recognised and given policy support from the European Union and member states [5].

One way to translate the technical benefits of shared PVs into reduced electricity costs for households is through use of an embedded network (EN). An EN is connected to the main distribution network through a ‘parent’ or ‘gateway’ meter, with individual customers connected to the EN through a ‘child’ or sub-meter. Although similar arrangements are referred to as building microgrids in some jurisdictions, an EN is differentiated from the common definition of a microgrid by being permanently connected to the main distribution network, without the ability to operate in island mode. An EN enables communities to aggregate their load and PV generation to participate in the retail market as a commercial sized buyer and thereby access reduced overall network charges and lower grid electricity rates. While buyers partnering with each other to get better deals in competitive markets might seem highly pro-market, it should be noted that ENs are not always seen in this light by policy makers, or by utilities for that matter. In Australia, for example, residential ENs have the dubious reputation of providing a vehicle for unscrupulous developers and Embedded Network Operators

(ENOs) to lock a captive consumer base into long-term electricity contracts with unfavourable tariff arrangements [6]. This has been reflected in approaches to EN regulation that facilitate consumers within an EN moving ‘on market’ to participate in the competitive electricity retail market [7]. Still, if it can be shown that ENs can deliver better outcomes for consumers than individual participation in retail electricity markets, then the case for supporting them is much clearer.

Previous research [8] has identified that, for many apartment buildings in Australia, sufficient value can be generated through this arrangement to significantly reduce aggregate electricity costs, while also supporting the case for PV deployment. However, there is a need to engage all apartment residents as energy consumers and all apartment owners as investors in PV and EN infrastructure, and this can present challenges. This provides an opportunity for the development of new and innovative business relationships between ENOs, apartment owners and residents¹ that enable fair distribution of the net benefits of ENs and rooftop PV generation. However, designing business models and allied tariff structures to ensure fair, efficient and effective allocation of the benefits, to be acceptable to all stakeholders, presents two challenges. It is necessary, firstly, to address split incentives between apartment owners and tenants and, secondly, to secure the participation of households with diverse energy needs. Indeed, while the benefits of load aggregation are in part due to diverse temporal distribution of electricity demand across different types of household within a building or community, this diversity can, paradoxically, hamper the equitable distribution of those benefits.

This paper describes a unique study of the potential distribution of costs and benefits between apartment owner-occupiers, tenants and landlords in an embedded network serving a 72-dwelling apartment complex with shared rooftop solar. As well as providing a case study to explore general approaches to RE cost-benefit distribution, the EN context is subject to particular constraints which affect the scope for balancing conflicting aims of effective cost recovery, perceived consumer fairness and overall economic efficiency.

The remainder of this section presents a brief overview of the pertinent academic literature which has informed the focus and

¹ Throughout this article, the terms ‘resident’ and ‘consumer’ refer to a person living in an apartment and responsible for paying the household electricity bill, whether they are an owner-occupier or rent-paying tenant, while the term ‘owner’ includes both owner-occupiers and landlords but excludes tenants.

methodological design of this study. Section 1.2 describes previous research relating to the deployment of PVs in apartment buildings. Section 1.3 summarises some key considerations in tariff design, including different approaches to fairness, equity, efficiency and consumer acceptability. Section 1.4 introduces some regulatory constraints affecting tariff design in the specific context of embedded networks. Finally, the opportunity presented by the gaps in the literature is presented in Section 1.5, along with an outline of the contribution made by this article.

1.2. Shared PVs in apartment buildings

Deployment of PVs on single-family housing has reached 25% in some countries, including Australia [2], but deployment on multi-family housing lags behind in all international jurisdictions [9], despite diverse regulatory environments [5]. Deployment of PVs in this housing sector has been hampered by multiple diverse and interrelated barriers [10], with varying impacts across different jurisdictions [11]. These include technical complexity, lack of profitability [12] and regulatory constraints. There is also a challenge in engaging a range of diverse stakeholders, as conflicting incentives between owner-occupiers, landlords and tenants can hinder investment in generation and distribution infrastructure [10].

In her analysis of reasons given by householders for not installing rooftop PVs, Zander [13] found that “coordination challenges with other parties” was the second highest barrier for apartment owners (after high installation cost), with “uncertainty about sharing costs and benefits” fourth highest, and “coordination challenges with landlord” and “fear of higher management/strata fees” the highest barriers for apartment renters. The paper highlights the necessity of establishing trust between apartment parties and the need for policy approaches to support fair sharing of costs and benefits.

One approach to distributing PV generation between apartments within a building is to allocate a proportion of roof space (and of the PV system) to each apartment, either on an equal basis or dependent on the amount or temporal distribution of energy consumption. In some apartment developments, an equal allocation of PV capacity takes the form of an individual PV system for each apartment, connected behind the consumer’s meter as is common in detached houses. However, higher self-consumption and self-sufficiency can be achieved by a single PV system with the generation shared between apartments [14] and increased by adding a shared battery [15–17]. Other technical arrangements are possible for distributing the generation from such a shared system [8]. However, use of an EN, where the RE generation is distributed within the building along with electricity imported from the grid, has the advantage of providing an opportunity to aggregate loads across the building and thereby leverage lower energy tariffs and reduced Distribution Network Use of Service (DUOS) charges for electricity imported from the grid.

Some studies have approached the distribution of energy within an EN by allocating a virtual share of the single PV system to each consumer and, in some cases [18, for example], allowing peer to peer (P2P) trading between consumers. Abada, Ehrenmann et al [19] use co-operative game theory to model such an arrangement and show that simple pro-rata rules for the allocation (per capita, pro-rata to volume or peak demand) can be efficient for similar households but, for households with diverse load profiles, there is a trade-off between equity and efficiency, and that increased efficiency also comes at the cost of increased complexity and, potentially, reduced consumer acceptance.

1.3. Key considerations in tariff design

A successful tariff enables the ‘effective’ cost recovery of operational and sunk costs involved in electricity provision, incentivising ‘efficient’ behaviours that increase net benefits for all consumers in aggregate, while also satisfying criteria of fairness and equity. The literature on

tariff design is extensive and testifies to the difficulty of achieving these aims; a comprehensive review is therefore beyond the scope of this article. This section is limited to a discussion of some diverse approaches to fairness, equity and consumer acceptability, and of tariff studies specific to embedded networks and renewable energy microgrids.

1.4. Tariff components

Electricity tariffs commonly include components applied to volumetric consumption (either at flat or time-varying rates) and/or peak demand, as well as fixed charges and conditional additions or reductions (such as late payment penalties or direct debit discounts). In broad terms, volumetric charges incentivise energy efficiency and load reduction, while ToU tariffs and demand (or capacity) charges incentivise load shifting and peak demand shaving.

Fully cost-reflective electricity pricing will charge a consumer according to the aggregated costs of their energy supply, which includes capital and operational costs for generating electricity as well as (sunk and future) capital and operational costs for the distribution network. For an EN with PVs, operational costs include metering, billing and administrative charges, as well as the commercial cost of electricity imported at the parent meter, which typically combines network charges (with relatively high peak demand charge and fixed components), a relatively low (ToU) wholesale volumetric charge, and retail margin. To be fully cost-reflective, an EN tariff is likely to reflect the structure of this parent tariff, while recovering all operational costs and repayments for EN and PV investment costs.

Fridgen et al [20] examined a range of tariff structures in residential microgrids and concluded that volumetric tariffs have a destabilizing effect that can be moderated by capacity charges and that “effective peak shaving is only possible in a co-operative setting.” Passey et al [21] show that, to be cost-reflective, network demand charges should be calculated on consumers’ coincident demand peaks, rather than their individual peak demand. Rieger et al [22] and Zhang et al [23] examined the use of a capacity tariff in conjunction with real-time volumetric pricing, in micro-grids with cooperative demand response and battery storage, and found that the capacity charge prevented demand response peaks and smoothed load profiles. These and many other studies point to the need for a level of complexity in tariff design but assume a high level of load flexibility and appliance automation (made possible by smart homes or smart grids) to facilitate optimal load shifting in response to price signals.

By contrast, a typical apartment building does not have a high level of co-operative demand response or significant levels of automation and its residents are not highly engaged in their energy management. Consumer interest in purchasing smart home energy management systems remains low [24]. The higher proportion of rental dwellings and lower average income in apartments relative to houses [25] may also reduce the ‘flexibility capital’ [26] of these consumers. In these circumstances, price efficiency may be less important in EN tariff design than consumer acceptability factors, including equity and transparency.

1.5. Fairness, equity, simplicity and transparency

Fairness and equity are widely seen as critical to consumer acceptance of tariffs, but their meaning is contested in the literature. Indeed, the meanings of cost-reflectivity, efficiency and fairness are sometimes, wrongly, conflated. Schittekatte [27] contrasts efficiency with fairness, and discusses the importance of achieving balance between the two.

Bunzl [28] and Hogan [29] argue that the classical economic assumption that fairness is delivered through efficiency or cost-reflectivity may be applicable when designing a system from scratch, but not when starting from a non-ideal (or unfair) state. Moreover, efficient tariffs rely on consumers having the ‘flexibility capital’ [26] to reduce or shift their demand in response to price signals and may therefore adversely affect consumers with inefficient homes or

appliances, or those whose energy use is constrained by lifestyle or health factors. Rather than being fair, efficient tariffs may, therefore, have the effect of magnifying pre-existing inequalities.

Brown et al [30] offer several possible interpretations of what fairness might mean, including one that emphasises the importance of gradualism, proposing that fairness in tariff design requires avoidance of drastic changes in consumer bills (commonly referred to as “bill shock”) in response to a tariff change. In the case of an EN where participation is optional, as is the case in Australia [31], this preference may become a hard constraint as any sharp increase in electricity bill when moving to the EN is likely to inhibit participation in, and thereby undermine sustainability of, the EN.

A considerable body of research supports energy consumers’ preference for simplicity in energy pricing. 30–40% of Australian energy users have never switched energy retailers or tariffs [24], and consumers report being “consistently confused and overwhelmed” by existing tariff structures and electricity bills, preferring simplicity and ease of management [32]. In February 2020, only 21% of consumers in New South Wales (NSW) had the smart or interval meters needed for ToU tariffs and only 35% of these had any type of cost-reflective tariff, so approximately 93% of consumers remain on flat tariffs [33] despite the plethora of available market offers. In Australia, as elsewhere, there is increasing recognition of a longstanding deficit in consumer trust across the electricity industry and understanding of the importance of transparency in addressing that deficit.

Nicolson et al [34] showed that loss-aversion (attaching greater importance to potential losses than potential savings) is highly significant in consumer reluctance to switch tariffs. Reassuring consumers that EN participation will not result in cost increases relies on their understanding of tariff arrangements and is likely to be undermined by complexity or opacity. Given the historic problems associated with ENs described above [6], the need for transparency is at least as vital here as elsewhere in the industry.

More broadly, fairness can be seen as an overarching term that encompasses “distributional issues (inflexibility, affordability and non-discrimination), transparency (simple and predictable) and last but not least, graduality” [35] and may be used as a proxy for public acceptability. Given the lack of consensus on the meaning of “fairness”, the term has been avoided in the analysis below. Instead, the more specific terms “savings equity”, “customer acceptance” and “universal benefit” (indicating that no customer has an increased bill) are used to assess the outcomes of applying different tariffs.

1.6. Regulatory constraints

Most Australian apartments are governed under strata title, similar to condominiums in the USA, whereby the internal wall surfaces of apartments belong to individual owners, but the building structure (including the roof) is owned collectively through the strata body [36] which manages building maintenance and improvements.

However, the legislated collective responsibility of strata owners does not always align with their real behaviours [37], and investment in infrastructure, including PVs or ENs, can be hampered by the conflicting agendas of multiple players - owner-occupiers, landlords, tenants, voluntary management committees and (often external) managing agents [38]. In particular, individual use of shared parts of the property, such as the roof space, requires a ‘strata bylaw’ as well as an agreement for allocation of the resource between apartments. In this context, deployment of a shared system on the shared roof space, for the benefit of all owners and residents may be more equitable, and easier to achieve,

than division of the resource.

A second regulatory consideration relates to electricity selling and network management, whereby consumers within an embedded network have the right to opt-out and purchase their electricity on the competitive retail market [31].

As a consequence of these constraints, deployment of PVs through an EN on an apartment building requires incentivisation of all apartment owners (the investment decision makers) and engagement of all apartment residents (the electricity consumers). In turn, this provides constraints on EN tariff design that, as well as providing a return on investment (ROI), tariffs must be attractive to *all* EN consumers to avoid risking collapse.

1.7. Contribution of this study

Electricity tariff design “requires tradeoffs, making it as much an art as a science” [39] and a tariff designed for the particular case of an EN with PVs must satisfy multiple criteria. It should recover costs and provide a financial return to owners in order to secure investment in EN and PV infrastructure; it should be cost-reflective, incentivising load reduction, peak demand shaving and solar self-consumption in order to reduce costs for all participants; it must be attractive to all its consumers, which requires transparency and simplicity as well as fairness; and it must leave no consumer worse off than under Business as Usual (BAU).

Thus, we propose a novel framework, a ‘quadrilemma’ of four conflicting requirements: cost-recovery, efficiency, consumer acceptance and universal benefit, which must be effectively addressed by EN tariff design. In particular, the need to provide universal benefit may directly conflict with the potential to incentivise particular desirable behaviours.

This study examines the effect of these constraints on the distribution of costs and benefits within a proposed EN with rooftop PVs in a real apartment complex, and explores the ability of different tariff structures to satisfy the quadrilemma. The combined benefit of retrofitting the building with an EN and PVs, compared to BAU and to other options for PV installation, is first calculated. Secondly, different approaches for distribution of this benefit, including directly to apartment *owners* through reduced strata charges or to *residents* through reduced electricity bills, are explored. The impact of different tariff arrangements (including Flat, ToU, Solar ToU and demand charge components) on the owners’ ROI, on residents’ bills, on equity and on incentives to reduce consumption, maximise solar self-consumption and reduce demand peaks is explored.

Other studies have examined the aggregated benefits for all stakeholders [8], determined optimum PV system sizes under specific market conditions [40], or simulated renewable energy trading between consumers with individual PV systems [18]. By contrast, this paper presents an exploration of a range of financial settings for distributing costs and benefits between stakeholders, given the technical and economic benefits of a shared PV system. Such cost-benefit distribution, under the particular set of regulatory and market constraints that affect the EN scenario, has not been previously explored in the literature.

Although aspects of the model used in this study have been used in previous studies [8,15], the method and analysis have here been developed to include a wider range of tariff structures and to include analysis of financial outcomes at the level of individual apartments, rather than aggregated across the building. The model is here applied to a case study of a 72-apartment complex, using half hourly, apartment-level load data collected over a year and PV generation data modelled using real weather data and the actual roof area and form of the complex. This serves two purposes. Firstly, it enables real-world validation

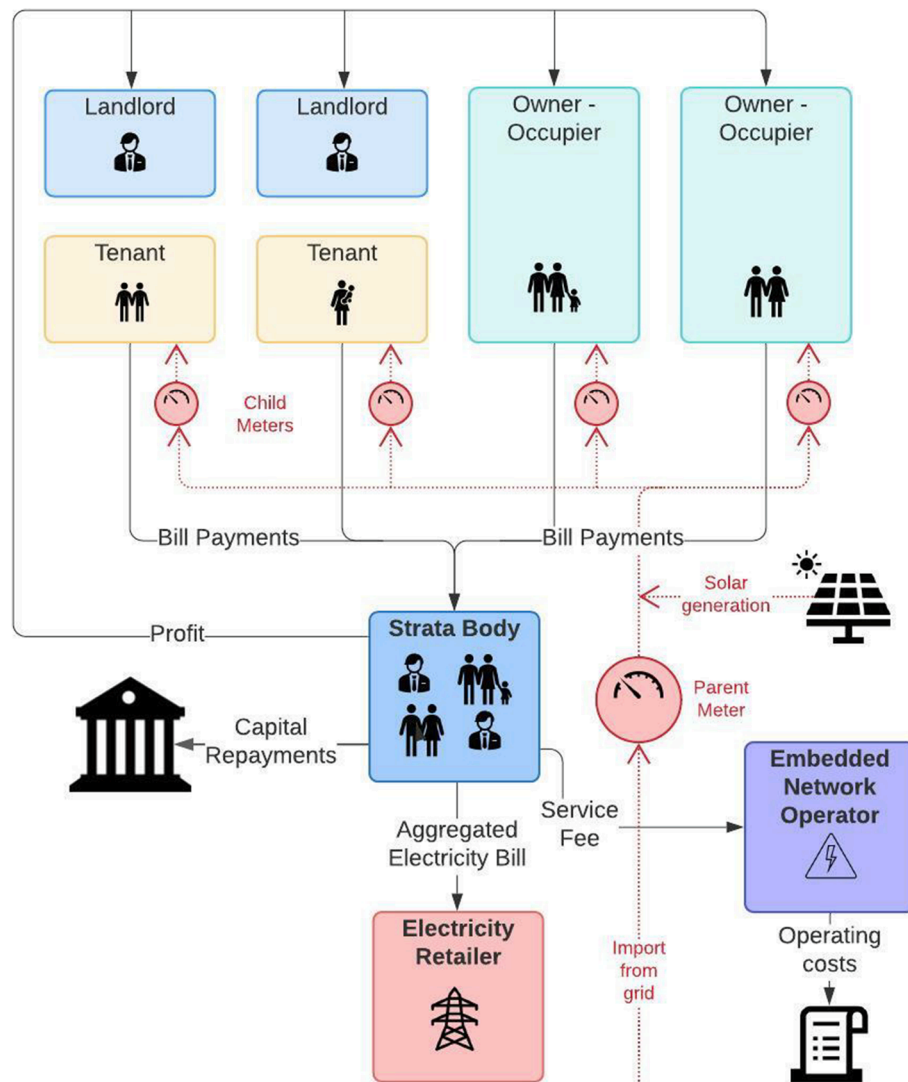


Fig. 1. Energy flows (dotted red line) and financial flows (solid black line) for an embedded network with photovoltaics, with only 4 apartments shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of the previous finding that deploying shared PVs distributed through an EN maximises the total financial benefit compared to deployment of PVs for individual apartments. Secondly, and more importantly, it enables a deeper analysis of the differential outcomes between different households and of the dependence of bill savings on the specific characteristics of apartment load profiles.

The contribution of this study is threefold. Firstly, it provides a unique analysis of the potential distribution of costs and benefits between different actors – owner-occupiers, landlord owners and tenants – in a real apartment complex, and the implications for households with diverse electricity consumption profiles. Secondly, a novel framework is applied to the case study, the ‘quadrilemma’ of cost-recovery, efficiency, consumer acceptance and universal benefit. This framework is used to analyse the trade-offs between the different aims of EN tariff design. Thirdly, it explores the impact of the particular regulatory and organisational constraints placed on embedded networks within a competitive retail market. In particular, whether the need to engage all stakeholders to avoid collapse of the EN arrangement effectively excludes a range of

potential tariff structures.

To the authors’ knowledge, none of these three contributions can be found in the existing literature. Firstly, distribution of the financial benefits of shared onsite renewable generation has not previously considered both the conflicting incentives of tenants and owners and the impacts on different households with diverse load profiles. Secondly, although customer preference for equity and transparency in tariff design is well established in the literature, the need for universal benefit or ‘no customer worse off’ has not been previously included as a constraint on tariff design in an energy community with diverse load profiles. Thirdly, the existing energy community literature has not addressed the impact of allowing customers to opt out of an energy community on the provision of cost-reflective and efficient tariff structures. The novelty of this research therefore lies in its consideration of these three elements, individually and in combination.

This leads to the finding that implementation of cost-reflective tariffs within an EN, to incentivise energy behaviours that could generate benefits for all stakeholders, cannot be reconciled with the need to

Table 1

Business as Usual tariffs (AU cents inclusive of GST). Discounts are applied to all fixed and volumetric components.

Tariff	Code	Type	Fixed Charge (c/day)	Peak (c/kWh)	Shoulder (c/kWh)	Off-Peak (c/kWh)	Discount %
BAU Flat	BAU_FL_15	Flat	85.00	30.00	30.00	30.00	15.0
BAU ToU	BAU_TOU_15	ToU	105.36	38.97	26.11	20.19	15.0

ensure that no stakeholder is worse off under the EN arrangement. This original finding has important policy implications and highlights potential limitations to individual market participation in achieving co-ordinated energy management. This study therefore makes an important contribution to understanding of a key challenge to increasing PV deployment in this potentially significant, but underdeveloped, market sector.

The findings of this research serve to illustrate the tensions inherent in tariff design, as well as highlighting the implications of competitive market principles for co-ordinated approaches to PV deployment, including in renewable energy communities. Although based on an Australian EN case study, the findings therefore have wider application.

The remainder of this article is arranged as follows. Section 2 describes the modelling method and the range of options explored, which includes tariff structures with fixed and time of use (ToU) volumetric and demand charges. Section 3 presents the particular case study used for the research and describes the associated datasets as well as the financial settings applied. The results are presented in Section 4, including financial outcomes for apartment residents and owners under different arrangements and a comparative analysis of different tariff structures in terms of their impact on the quadrilemma described above. Finally, in Section 5, the implications for EN tariff design and policy are discussed, and some suggestions made for further research in this area.

2. Method and tariff scenarios

2.1. Method

This study uses the open source *Multi-Occupancy Residential Electricity with PVs and Storage* (morePVs) tool [41], which models electricity flows and financial transactions in apartment buildings with PVs. The operation of the tool was verified by making a series of incremental changes to the input data across a range of both simplified and realistic scenarios, with the results compared to those obtained by manual calculation. The original tool has been used for two previous published studies [8,15]. However, for this study, it has been developed, extended and reverified. New functionality includes modelling of a wider range of residential tariffs, including those with a demand charge component, and provision of apartment-level outputs.

The model is applied to the case study described in Section 3. Firstly, a comparison is made between the aggregate cost savings generated across the complex by installing PVs for individual apartments behind-the-meter (*BTM_I*), by aggregating all building loads in an embedded network (*EN*) and by combining an EN with a shared PV system (*EN_PV*). Secondly, for the *EN_PV* arrangement, different tariff arrangements for distributing costs and benefits between apartment owners and residents are analysed to explore the diversity of outcomes between different households.

2.2. Embedded network business models

For the EN arrangement, a number of business models are possible. The EN may be owned and operated by a third party ENO which is responsible for EN and PV investment costs, may lease the roofspace from the strata body, and returns profits to its shareholders. Alternatively, the EN may be owned by the strata body, with operation, metering and billing outsourced to an ENO in return for a service fee (which includes an appropriate margin) and EN profits able to be returned to apartment owners through reduced strata fees. Other

potential arrangements involve sharing the profits between strata body and the ENO.

In this study, the second option is assumed, in order to examine distribution of maximum possible benefits to apartment owners and residents. The investment costs of the EN and PVs are paid by the strata body, whether the money is borrowed, paid from reserves or raised through a levy on owners. These costs may be shared between all the apartment owners in proportion to owners' 'unit entitlements' (roughly correlated with the relative value of their apartment) or equally. For simplicity and transparency, an equal share is assumed for the purposes of this study.

Fig. 1 illustrates the energy and financial flows under this arrangement. The strata body purchases electricity, measured at the parent meter, from an electricity retailer, supplements it with onsite solar generation, and sells it to apartment residents (tenants and owner-occupiers), measured at the child meters. After paying operating costs (including metering, billing and an appropriate margin) to the ENO, and repayments for capital investment costs, remaining profits are distributed equally between all apartment owners (landlords and owner-occupiers).

Under this arrangement, the total benefits derived from deployment of the EN and PV (after capital repayments and operating costs) can be distributed to apartment owners and residents. One edge case is to distribute the benefits equally to all apartment owners; the other is for the EN to operate on a non-profit basis, distributing all the benefits as bill savings for residents. In between these two extremes, the benefits can be shared between owners and residents.

2.3. External electricity tariffs

2.3.1. BAU tariffs

Household electricity bills under the EN are compared with BAU scenarios in which households pay an on-market Flat or ToU² tariff (Table 1), equivalent to 15% discount on all components of the regulated "Default Market Offer" (DMO) tariffs; this is typical of the tariff paid by NSW residential consumers³.

2.3.2. Parent tariffs

The aggregation of building loads in an EN enables the strata body / ENO to access a commercial tariff at the parent meter, comprising a regulated network component (determined according to voltage at the connection point and annual load) and a negotiated wholesale and retail component. Typically, the volumetric component is much lower than for a residential tariff, while a capacity or demand charge is included, based on the annual peak consumption. Table 2 shows the parent tariff used for the modelling, with regulated network and environmental charges, wholesale volumetric components based on the average NSW wholesale price between Jan 2018 and Jan 2020, and a typical retail margin [43].

² The modelling uses Ausgrid spring-summer-autumn ToU periods: peak period is 14:00–20:00 on weekdays; shoulder period is 7:00–14:00 and 20:00–22:00 weekdays and off-peak is 22:00–7:00 weekdays and all weekend. The shorter winter peak period (17:00–21:00) which applies for three months June to August was not modelled.

³ BAU ToU tariff uses the average of the peak, shoulder and off-peak prices of Enova, Powershop, Origin and LPE from comparison website Energy Made Easy [42], while the flat tariff is Origin Max Saver Online Special.

Table 2
Parent Tariff (AU cents inclusive of GST).

Component	Fixed Charge (c/day)	Demand Charge (c/kW/day)	Peak (c/kWh)	Shoulder (c/kWh)	Off-Peak (c/kWh)
Network charges (Ausgrid EA305) [44]	1817.60	36.09	6.81	2.50	1.23
Wholesale			11.91	9.65	8.18
Environmental Charge [45]			1.64	1.64	1.64
Retail Margin	181.76	3.61	2.04	1.38	1.11
Parent Tariff	1999.36	39.70	20.76	13.53	10.52

Table 3
Internal Embedded Network Tariffs (AU cents inclusive of GST).

Tariff	Code	Type	Fixed Charge (c/day)	Demand Charge (c/kW/day)	Peak (c/kWh)	Shoulder (c/kWh)	Off-Peak (c/kWh)	Discount %
Flat Rate	FL_15	Flat	85.00	–	30.00			15.0
	FL_H	Flat	85.00	–	30.00			23.5
	FL_Z	Flat	85.00	–	30.00			31.9
ToU	TOU_15	ToU	105.36	–	38.97	26.11	20.19	15.0
	TOU_H	ToU	105.36	–	38.97	26.11	20.19	21.5
	TOU_Z	ToU	105.36	–	38.97	26.11	20.19	28.0
Solar ToU	STOU_15	SToU	105.36	–	38.97	26.11	20.19	15.0
	STOU_H	SToU	105.36	–	38.97	26.11	20.19	19.5
	STOU_Z	SToU	105.36	–	38.97	26.11	20.19	26.0
Pass Through (Demand at consumer peak)	PT_D	PT	27.77	39.70	20.76	13.53	10.52	–
	PT_N	PT	27.77	30.70	20.76	13.53	10.52	–
	PT_Z	PT	27.77	20.95	20.76	13.53	10.52	–
Pass Through (Demand at building peak)	PTBP_D	PTBP	27.77	39.70	20.76	13.53	10.52	–
	PTBP_Z	PTBP	27.77	70.10	20.76	13.53	10.52	–

2.3.3. EN tariffs

A variety of tariff structures and rates are modelled for the distribution of costs and benefits within the EN, as summarised in Table 3.

The basic Flat and ToU tariffs use the same underlying structure and rates as the BAU tariffs (Section 2.3.1), with a discount of 15% or higher to ensure that no consumers pay more than BAU. A variation on the ToU tariff structure, Solar ToU (SToU), uses an additional weekday off-peak period of 10:00 – 14:00 to incentivise load-shifting to maximise solar self-consumption. Additional discounts are applied to determine the appropriate discount level to give income neutrality (after operating costs and margin described above) for the strata body / ENO with no discount applied to common property bills; in this scenario (Z), all the benefit of EN and PVs is passed to residents through discounted electricity bills. Additionally, the discount required to pass half the total benefit to consumers (H) is determined and applied to all three volumetric tariff structures. Thus, discount rates have been selected such that _15 tariffs result in the maximum possible average bill (while still enabling customer retention), _Z tariffs result in the minimum possible average bill (while avoiding the EN operating at loss), and _H tariffs sit at the midpoint of these two edge cases.

An alternative EN tariff structure passes through the commercial tariff structure seen at the parent meter (Table 2). This pass-through (PT) tariff uses the same rates as the parent tariff but with the fixed daily connection charge shared equally between all 72 apartments in the building. This has been tested with a range of demand charge rates: equal to the parent tariff (PT_D), decreased to reduce EN profit to zero (PT_Z), or set to ensure the average consumer bill is the same as for BAU (PT_N). Finally, a more cost-reflective version of the pass-through tariff (PTBP) is tested, with the demand charge component applied to the consumer's contribution to the building annual peak demand, rather than to the consumer's own peak, with parent tariff rates passed through directly (PTBP_D) or set to generate zero EN profit (PTBP_Z).

3. Case study

3.1. Load profiles

The study uses data from a real 72-apartment complex in northern Sydney, NSW. 30-minute load data was collected over the course of a year (June 2016 to May 2017) from 28 of the apartments, the common property load and the whole building load, using Wattwatchers Auditor devices [46]. The complex comprises three 3-storey buildings which house a diverse population including single households, working couples, young families and retirees. Ethical and privacy considerations required individual permissions from residents to collect the load data, and this was granted by 28 households. Demographic and household characteristics for a subset of these apartments were collected via an online survey. The electrical arrangement of the complex is identical to that of a single 72-apartment building and, for simplicity, it is treated here as a single building where “building” refers to the 3-building complex.

Although the apartments without measured load data are not included in the analysis of household costs and benefits, it is necessary to estimate their contribution ($E_{u,i}$) to the total building load ($E_{building}$) profile and to the income of the strata body / ENO. For this purpose, the residual building load (i.e., total building load minus the common property (CP) and 28 metered apartment loads) is first divided equally between the 44 unmetered apartments, as shown in (1), and then a peak demand profile ($E_{p,i}$) is added. Here, $E_{m,j}$ is the load profile for the j th metered apartment, E_{cp} is the load profile for the common property, and all load profiles comprise 17,520 half-hourly energy values.

$$E_{u,i} = \frac{E_{building} - E_{cp} - \sum_{j=1}^{28} E_{m,j}}{44} + E_{p,i} | i \in (1, 2, \dots, 44) \quad (1)$$

The peak demand profiles ($E_{p,i}$) for unknown households comprise a

Table 4
Photovoltaic allocation for individual behind the meter arrangement.

Total PV capacity (kW)	Common Property PV capacity (kW)	Apartment PV Capacity (kW)
75	16.2	0.82
110	23.7	1.20
145	31.2	1.58

single peak energy value $E_{peak,i}$ for timestamp $t_{peak,i}$ and 17,519 zero values, and are based on the distribution of peak demand in the data for known households. The diversity of load profiles and household characteristics across the metered apartments does not indicate any obvious systemic bias in types of household and, therefore, the assumption that the distribution of peak demand in the metered apartments is representative of the whole complex is a reasonable, though imperfect, assumption.

The magnitude and timestamp of peak demand (within the demand charging period of the parent tariff) for each metered apartment is found and ranked in size order. The sizes of the annual peak demand ($E_{peak,i}$) for the unmetered apartments are distributed to align with the distribution of the metered apartments, while their timestamps ($t_{peak,i}$) are selected to be the same times as those of the metered apartments but shifted by seven or fourteen days to avoid artificially increased coincidence of peak demand.

To characterise the load profiles of each apartment, three metrics are used: Average daily load (kWh/day), annual peak demand (kW), and a 'self-consumption metric', measuring alignment of the load with PV generation, $M_{self-consumption}$. The latter is calculated using (2), where $\mathbf{p}$ is the normalised annual PV generation profile and $\mathbf{e}$ is the normalised annual load profile, both expressed as column vectors, and $\mathbf{p} \cdot \mathbf{e}$ indicates

the scalar product of the two vectors.

$$M_{self-consumption} = \mathbf{p} \cdot \mathbf{e} / \mathbf{p} \cdot \mathbf{p} \times 100\% \quad (2)$$

3.2. PV profiles

Three possible PV system sizes are analysed, designed to fit the unshaded roof area of the complex, using aerial imagery [47] to identify obstructions. The largest system utilises all unshaded North-, East- and West-facing roof planes, resulting in a dc system size of 145 kW. The energy generated by this system is simulated using the PV Watts tool [48] in NREL's System Advisor Model [49] using Australian Bureau of Meteorology (BOM) satellite-derived irradiance data [50] and temperature and wind-speed for the same period as the load data (June 2016 to May 2017) from the nearest BOM Automatic Weather Station. Two smaller systems with dc sizes of 75 kW and 110 kW are modelled by scaling the output of the larger system.

For the *EN* PV arrangement, the system is connected between the parent and child meters of the EN. For *BTM*, the total PV generation is divided into 73 smaller systems, each connected behind an individual consumer meter, as follows: the CP load is connected to a system sized proportional to the CP share of the total building load (21.5%) and the remaining PV capacity is divided equally amongst the apartments, as shown in Table 4.

3.3. Capital and operating costs

The capital expenditure (capex) for each PV system has been estimated, based on the average installation cost of commercial PV systems of different sizes in NSW (AU\$1.07/W for 75 kW, \$0.93/W for 110 kW

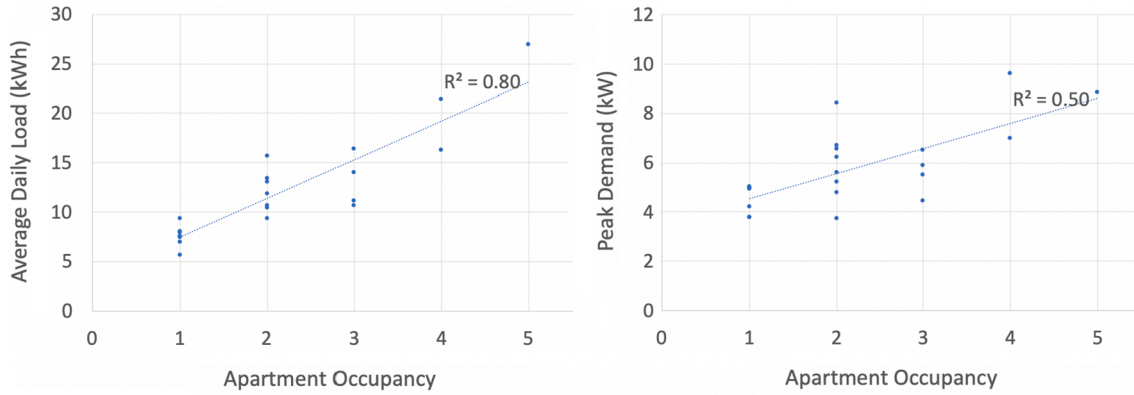


Fig. 2. Apartment load (left) and peak demand (right) relative to occupancy.

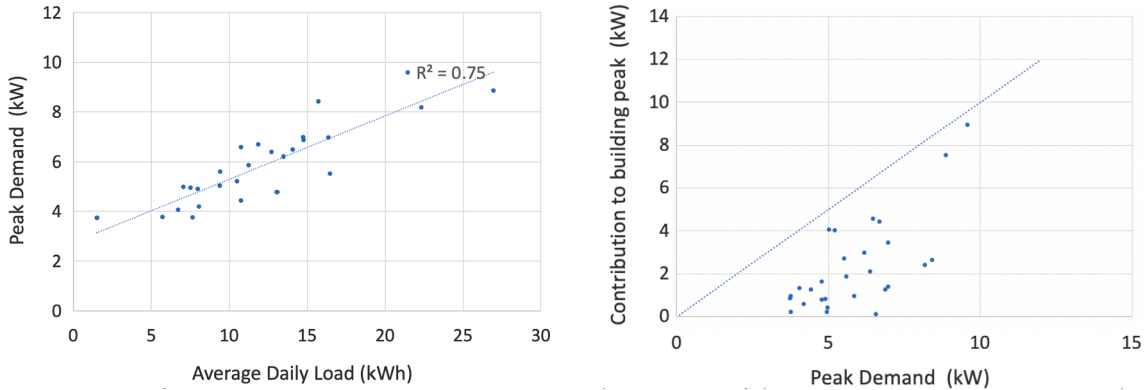


Fig. 3. Household peak demand (between 2 pm and 8 pm on weekdays) relative to average daily load (left) and household contribution to building peak relative to household peak demand (right). The dotted lines show a line of best fit (left) and line of equality (right).

Table 5

Photovoltaic self-consumption and aggregate building benefit for different photovoltaic system configurations.

PV configuration and capacity	PV self-consumption	20-year NPV of aggregate building savings (AU\$ per apartment)
BAU	–	–
BTM_I 75 kW	81%	\$1,321
BTM_I 110 kW	69%	\$1,624
BTM_I 145 kW	51%	\$1,481
EN, no PVs	–	\$2,231
EN_PV 75 kW	93%	\$2,430
EN_PV 110 kW	80%	\$2,436
EN_PV 145 kW	68%	\$2,073

Table 6

Sensitivity of 20-year net present value of aggregate building savings (per apartment) to photovoltaic system size and increased photovoltaic capex.

System Size	Additional Capex		
	0%	25%	50%
75 kW	\$3,098	\$2,764	\$2,430
110 kW	\$3,287	\$2,861	\$2,436
145 kW	\$3,158	\$2,615	\$2,073

and \$0.90/W for 145 kW [51]. Cost of inverter replacement (after ten years), at 20% of total PV installation cost, is also included. Anecdotal data [52] suggests installation costs are increased by up to 50% per installed Watt for apartment buildings, to account for safe access costs for a 3-storey sloped roof as well as additional grid protection requirements for systems above 30 kW [10]. Sensitivity of the results to increased costs of 0%, 25% and 50% is therefore tested (Section 4.2) and the additional 50% value is used as a conservative cost estimate for the analysis in Section 4.3.

Capex for retrofitting an EN to the building is dependent on the level of upgrade required to existing switchboards and wiring and is therefore highly building-specific. A value of AU\$20,000 has been used, based on the medium scenario used in previous modelling [8], additional to a cost of AU\$400 per apartment for meter replacement. Operating expenditure (opex) for the EN, including charges for metering and billing as well as an operating margin for the ENO, have been estimated at AU\$250 per consumer, in line with previous studies [8].

Note that all financial quantities are presented in Australian dollars, where AU\$ 1.00 is equivalent to EUR 0.6258 or US\$ 0.7356, as of July 2021 [53].

4. Results and discussion

4.1. Analysis of apartment load profiles

This section describes some characteristics of the load profiles, including the relationship between load and demand metrics, which drive tariff impacts, and occupancy.

4.2. Relationship between energy use and household characteristics

Fig. 2 shows the dependence of daily load (the key driver of volumetric tariff impacts) and annual peak demand on the number of apartment occupants for all metered apartments. Note that the correlation of occupancy with average daily load (coefficient of determination, $R^2 = 0.80$) is higher than for winter or summer load (0.75 and 0.77, respectively). Unsurprisingly, the annual demand peak (between 2 pm and 8 pm on weekdays) also increases with occupancy but the correlation is low. In general, this peak demand (which drives demand charges) increases with average daily load (Fig. 3, left). Importantly (Fig. 3, right), household peak demand is consistently higher than, and shows

Table 7

Reduced Year 1 consumer bills and owner benefits for Flat, Time of Use and Solar Time of Use tariffs for an embedded network with 75 kW photovoltaics.

Tariff Name	Tariff Type & Discount	Average Apartment Bill Savings (1st Year)	Owner benefit (1st year)
FL_15	Flat 15.0%	\$-	\$279
FL_H	Flat 23.5%	\$139	\$139
FL_Z	Flat 31.9%	\$279	\$-
TOU_15	ToU 15.0%	\$-	\$209
TOU_H	ToU 21.5%	\$105	\$105
TOU_Z	ToU 28.0%	\$209	\$-
STOU_15	SToU 15.0%	\$35	\$175
STOU_H	SToU 19.5%	\$105	\$105
STOU_Z	SToU 26.0%	\$209	\$-

little correlation with, household contribution to the building peak (which drives the demand charge levied at the parent meter of an EN).

4.3. Comparison of different installation arrangements

In this section, the benefit of different PV systems and installation arrangements – individual behind the meter and embedded network – is explored. Table 5 shows the PV self-consumption under different arrangements, along with the 20-year NPV of savings (relative to BAU), aggregated across the whole building and normalised for the number of apartments. The latter represents the available savings that can be distributed between owners and residents.

Table 5 shows that the dual benefits of high self-consumption and access to commercial tariffs make EN_PV the most financially advantageous arrangement for the building as a whole. This aligns with previous studies based on virtual buildings [8]. For both arrangements, savings are greater for a 110 kW system than a 75 kW system due to the greater generation and lower (\$/W) capex costs of the larger system, but the 145 kW system creates lower savings than 110 kW due to its lower self-consumption and similar capex. However, for EN_PV, there is very little difference in outcome between 75 kW and 110 kW systems. Table 6 shows the sensitivity of total building benefit (expressed as 20-year NPV per apartment) for EN_PV of different system sizes to increased installation costs for PVs on apartment buildings (Section 3.3), the capex premium. Note that, with lower capex costs, the benefit of the larger 110 kW system would be more significant.

4.4. Cost/benefit distribution: Internal tariffs

Given the total building benefits described above for EN_PV, an analysis of their distribution between apartment owners and residents is now presented. For this analysis, the + 50% estimate of PV capex cost is used to give a conservative estimate of the total building benefits and to align with anecdotal evidence discussed in Section 3.3, and the smaller 75 kW system size has been chosen due to the only marginal benefits of larger sizing in this cost scenario and its lower capex.

4.5. Benefits shared equally between apartment owners

As discussed in Section 2.2, one option is to distribute all the benefits to owners on an equal basis, which can be achieved by using the BAU tariff structures and rates within the EN, that is DMO Flat and ToU tariffs with 15% discount applied to all components. Residents' electricity bills are thereby unchanged and the benefits are accrued by the strata body which can pass them on to owners through reduced strata fees and/or reduced common property electricity bills. If costs and benefits are distributed equally, after payment of operating costs and capex repayments, this results in an annual benefit of AU\$279 per apartment owner using the BAU_Flat tariff, and AU\$209 using BAU_TOU.

This arrangement ensures equal distribution of benefits in return for equal investment, and owner-occupiers are incentivised to participate in

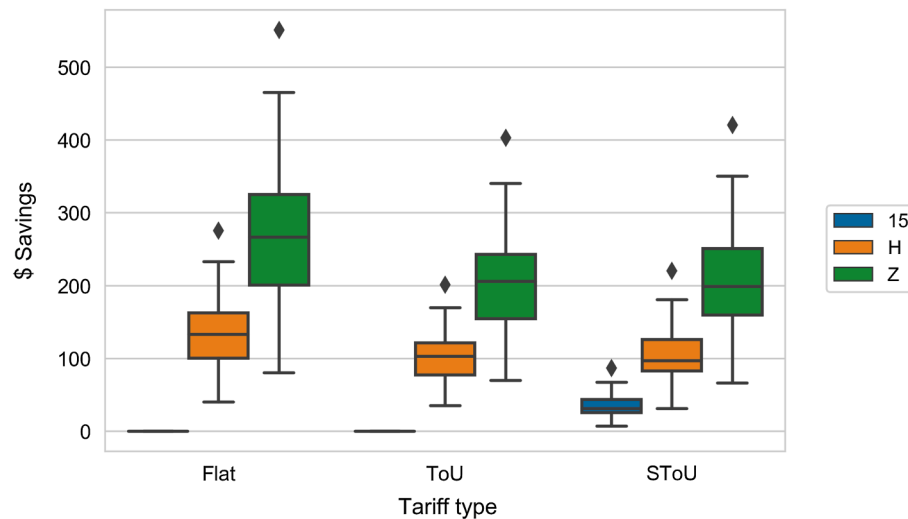


Fig. 4. Year 1 savings for Flat, Time of Use and Solar Time of Use tariffs with different levels of discounts off all tariff components.

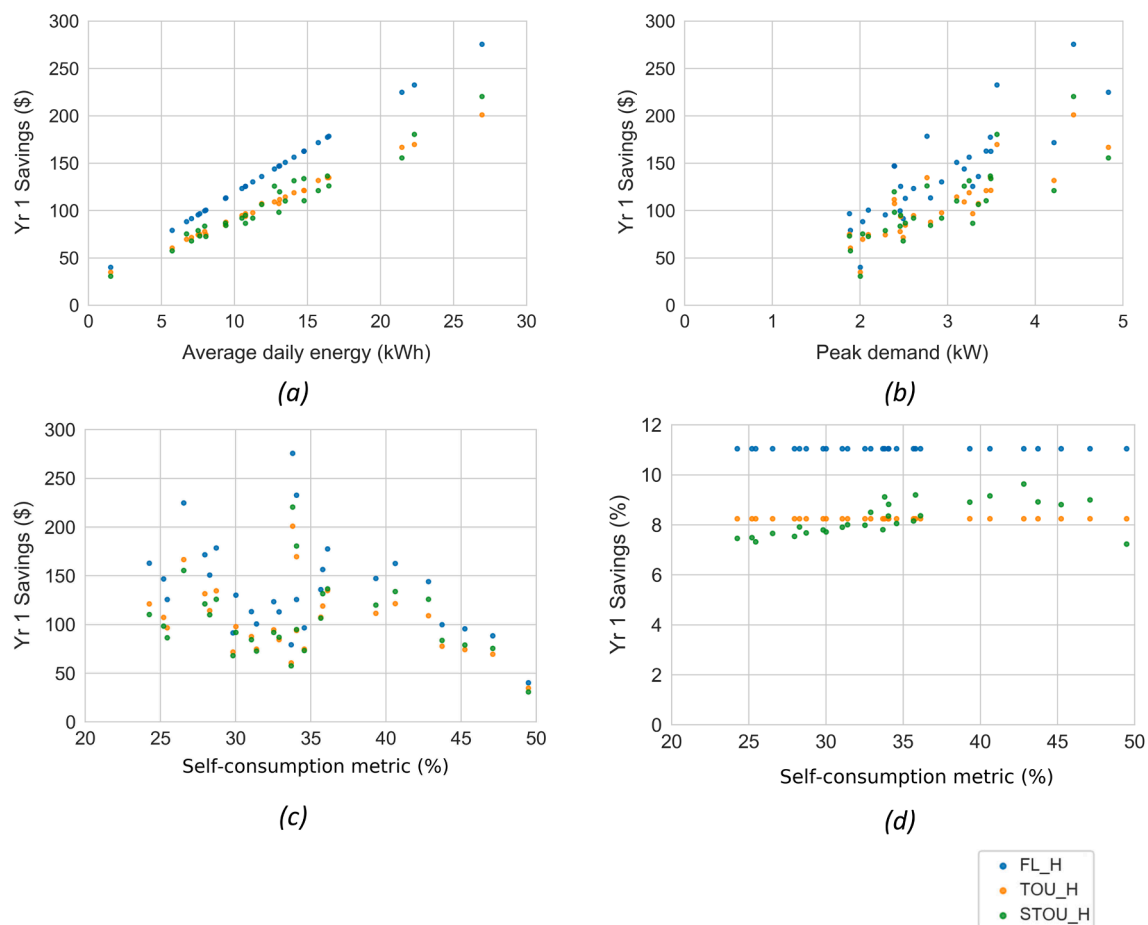


Fig. 5. Year 1 consumer savings related to (a) average daily energy, (b) peak demand, (c) self-consumption metric and (d) as a percentage relative to self-consumption metric for flat (FL_H), time of use (TOU_H) and solar time of use (STOU_H) tariffs.

the EN by the ROI. However, for tenanted apartments, no benefit is passed directly to the tenants who may therefore prefer to seek competitive offers on-market rather than participate in the EN. Although landlords could pass the benefit to tenants through reduced rent, they cannot require the tenant to buy their energy from the EN. As an electricity consumer, the tenant has the right to leave the EN and move on-

market under energy retail legislation, while the landlord is specifically prohibited from requiring a tenant to use a specific utility provider under property rental law [54]. This lack of benefit to tenants could result in reduced participation and therefore reduced income for the EN which, in turn, reduces the incentive for owners to participate.

Table 8

Correlation (Pearson's coefficient) between average bill savings and load characteristics for flat, time of use and solar time of use tariffs.

	FL_H	TOU_H	STOU_H
\$ savings vs average daily energy	1.000	0.999	0.977
\$ savings vs peak demand	0.845	0.858	0.806
\$ savings vs self-consumption metric	-0.391	-0.384	-0.220
% savings vs self-consumption metric	-0.150	-0.153	0.592

4.6. Benefits shared between owners and residents through volumetric tariffs

In order to avoid the 'mini-death-spiral' described above, residents can be incentivised with lower bills than are typically available on-market, sharing the benefits of EN + PVs between tenants and owners.

Table 7 shows how the total \$279 benefit per apartment is shared, on average, between residents' bill savings and EN profits passed on to owners. The third column shows the *average* first year bill savings available to residents under Flat, ToU and SToU tariffs with different levels of discount. Note that bill savings for ToU and SToU are relative to BAU ToU, while those for Flat EN tariffs are relative to BAU Flat. With the 15% discounted Flat and ToU tariffs, apartment bills are the same as BAU and all the benefits of EN and PVs flow to apartment owners through reduced strata fees and/or common property bills as described in 4.3.1 above, whilst 15% discounted SToU generates a small bill saving due to the additional off-peak period, but the majority of the benefit is again passed to owners.

Conversely, *FL_Z*, *TOU_Z* and *STOU_Z* tariffs are discounted such that all the benefits of *EN_PV* are passed to residents as bill savings, leaving strata fees (and CP bills) unchanged. In this scenario, landlords receive

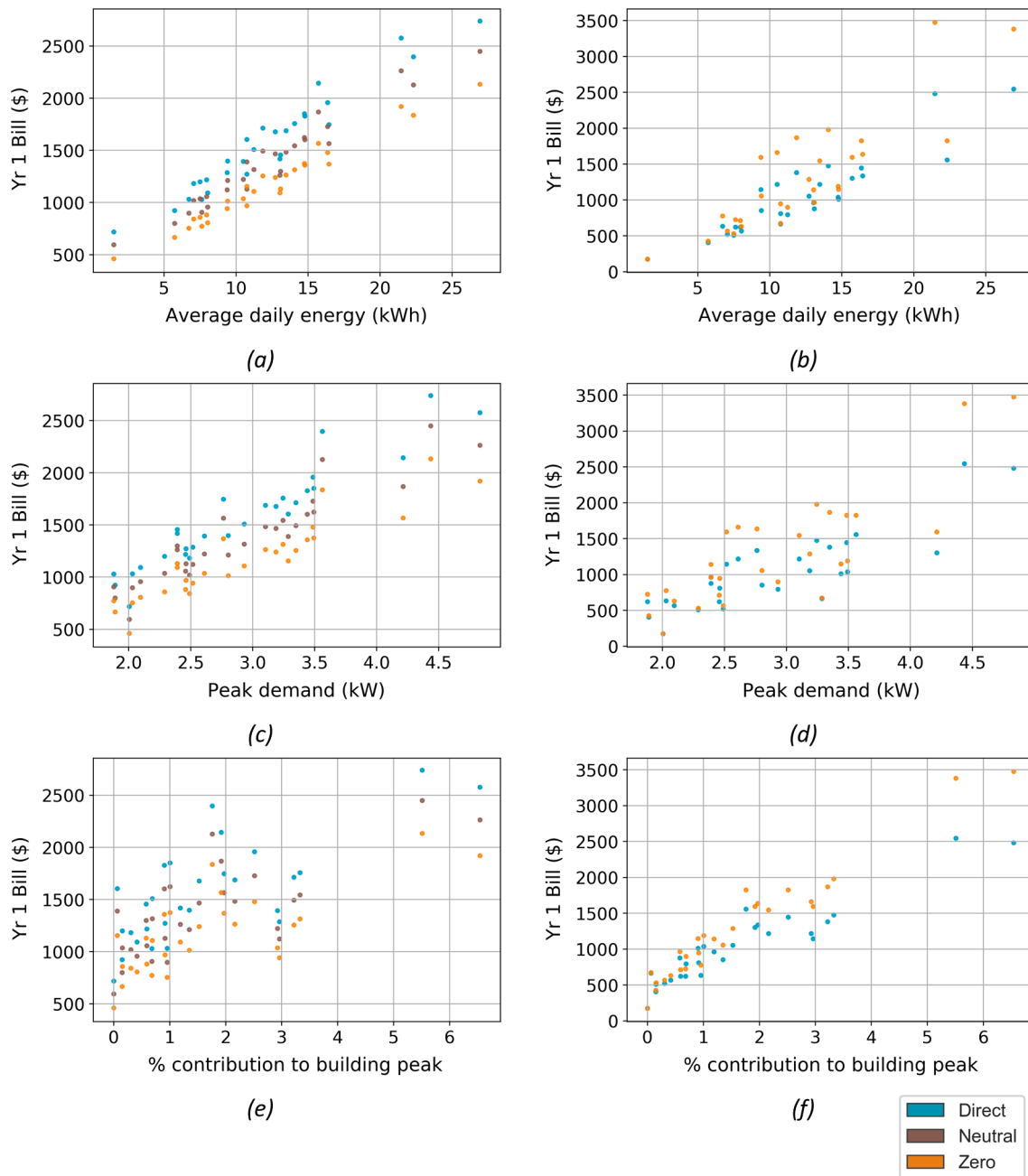


Fig. 6. Year 1 bills for pass through tariffs with different demand charge components applied to consumer peak (PT, left) or applied to consumer contribution to the building peak (PTBP, right), relative to average daily energy (top), peak consumer demand (centre), and consumer contribution to building peak (bottom).

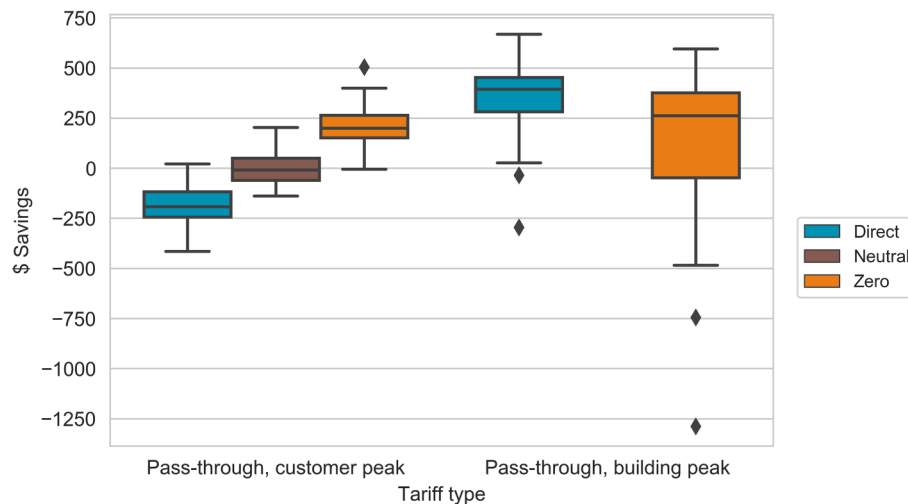


Fig. 7. Year 1 savings for pass through tariffs with different demand charge components applied to consumer peak (PT) or to consumer contribution to building peak (PTBP).

Table 9

Average year 1 consumer bill reductions and owner benefits for pass-through tariffs, with demand charges applied to consumer's peak and to consumer's contribution to building peak.

Tariff Name	Demand charge (c/kW/day)	Average Apartment Bill Savings (1st Year)	Owner benefit (1st year)
PT_D	39.70	-\$193	\$403
PT_N	30.70	\$-	\$209
PT_Z	20.95	\$209	\$-
PTBP_D	39.70	\$405	-\$195
PTBP_Z	70.10	\$209	\$-

no direct benefit, although the reduced bills could act as an incentive to attract tenants. *FL_H*, *TOU_H* and *STOU_H* represent the mid-point between these two edge cases, with benefits shared equally between residents and owners. Note that average benefits for owner-occupiers are the same in all cases as they are both owners and residents.

However, although the average bill savings shown in Table 7 are significant, they are distributed unevenly between residents, as shown in Fig. 4.

With discount rates higher than BAU, all consumers enjoy lower bills, with benefits shared less equally as discounts increase (Fig. 4). Due to their higher bills, consumers with high consumption, which commonly corresponds to higher peak demand (Fig. 3) and to larger households (Fig. 2), derive most benefit from moving to the EN (Fig. 5 (a) and (b)). Table 8 shows the high correlation of savings with average daily energy, and slightly lower correlation with peak demand. This impact dwarfs the small additional benefit accruing to consumers with high self-consumption metric under a STOU tariff (Fig. 5c), although the latter can be seen more clearly expressed as a percentage of bills (Fig. 5d), with a correlation coefficient of 0.59 (Table 8).

Note also that consumers' contributions to the peak building demand are not reflected in their bills under these tariffs, while the peak demand charge accounts for 29% of the commercial tariff payable at the parent meter.

4.7. Benefits shared between residents through pass-through tariffs

Passing through the tariff paid at the parent meter to residents is a more efficient way to ensure cost-reflectivity in consumer bills. This is illustrated by Fig. 6 which shows that, under all the pass-through tariff structures but particularly for PTBP tariffs, consumers with higher energy consumption, peak demand and contribution to the building peak

demand pay higher bills. However, Fig. 7 shows that the bill savings relative to BAU are highly variable and are negative for some consumers.

Despite substantially lower fixed and volumetric rates relative to the ToU BAU tariff, the direct pass-through tariff (PT_D) results in an average increase of \$193 in year 1 consumer bills due to the additional demand charge component, while the EN generates an average profit of \$403 per apartment, which flows through to apartment owners, as shown in Table 9. Moreover, the average bill changes conceal the impact on individual consumer bills which, as Fig. 7 illustrates, is highly variable, ranging from a saving of \$20 to a bill increase of \$416, which is clearly problematic for customer retention.

Reducing the demand component in the pass-through tariff to 30.70c/kW/day (PT_N) results in cost-neutral average bills, while the EN maintains profits of \$209 per apartment. However, while some consumers are better off, others are still paying up to \$139 more than BAU. A further reduction of the demand charge to 20.95c/kW/day (PT_Z) results in zero net EN profit and ensures bill savings (averaging \$209) for all consumers, except for one apartment with very low energy use (due to being empty for much of the year) which sees a modest bill increase of \$5. Although this PT_Z tariff passes all benefits to consumer bills and, with one minor exception, no consumer is worse off, the distribution of savings is very uneven, and is uncorrelated with energy use, peak demand, contribution to building peak or self-consumption metric.

If the direct pass-through tariff has the demand charge applied to the consumer contribution to the building peak (PTBP_D), the average bill saving is \$405 compared to BAU. However, the strata body / ENO is only recovering the cost of its energy bill, not repayments for EN and PV infrastructure, so the result is a net EN loss of \$195 to pass on to apartment owners. To recover these costs from resident bills, the demand charge component of the tariff must be more than doubled to 70.10c/kW/day (PTBP_Z). This gives the same average bill saving (\$209) as PT_Z, with greater correlation between consumers' bills and their contribution to building peak (Fig. 6(f)) and, therefore, greater cost-reflectivity. However, the impacts on individual bills are much more diverse (Fig. 7), ranging from a saving of \$595 to a bill increase of \$1288.

4.8. Impact of tariff structure on the quadrilemma

Table 10 summarises to what degree each of the tariff structures achieves the conflicting aims of cost-recovery (return on investment), efficiency, consumer acceptance and universal benefit. Note that the more cost-reflective pass-through tariffs achieve *neither* a positive return on investment *nor* universal consumer benefit.

Table 10
Summary of tariff impacts on desired outcomes.

Tariff Type	Tariff	Return on Investment	Efficiency / Incentivised behaviour		Solar self-consumption		Consumer acceptability		Universal benefit (no-one worse off)
			Reduced consumption	Peak demand shaving			Transparency	Savings equity	
Flat Rate	FL_H	Positive	✓	×	×	×	✓	(✓)	✓
Time of Use	TOU_H	Positive	✓	(✓)	×	×	✓	(✓)	✓
Solar Time of Use	STOU_H	Positive	✓	(✓)	(✓)	(✓)	✓	(✓)	✓
Pass-Through (consumer peak)	PT_Z	Neutral	(✓)	✓	×	×	×	×	×
Pass-Through (building peak)	PTBP_Z	Neutral	(✓)	✓	×	×	×	×	×
✓✓ strong positive, ✓ positive, (✓) slight positive, × negative, ×× strong negative.									

If BAU tariff arrangements (whether Flat or ToU) are replicated within the EN, benefits can be distributed equally to all apartment owners, maximising the ROI and fully incentivising them to implement the EN-PV installation. However, with no reduction in residents' bills, this arrangement provides no incentive to tenants to participate in the EN which is therefore likely to be unsustainable. Applying greater discounts to these tariff structures incentivises tenants to participate, while still providing ROI to apartment owners (with owner-occupiers receiving dual benefits), but the benefits are no longer shared equally between apartments, with greater savings accruing to higher energy users. However, these volumetric tariffs incentivise greater energy efficiency and, in the case of ToU, load shifting to off-peak periods, while STOU also provides a small incentive for solar self-consumption, so they deliver some degree of efficiency.

Greater cost-reflectivity can be achieved by passing through the parent tariff to EN consumers. However, if the demand charge is applied to households' contributions to the building peak, some residents will experience severe bill shock compared to BAU and the EN is unlikely to be sustainable. Additionally, the lack of transparency and bill certainty inherent in this structure is likely to make it unattractive even to some consumers who might benefit from it. If the demand charge is, instead, applied to household peaks, household demand management is made somewhat easier, but some consumers will still be significantly worse off than BAU, unless the demand charge component is reduced to a level where the EN makes no profit. The introduction of demand charges within the EN therefore makes it impossible to maintain incentives for both landlords and tenants.

5. Conclusion

This research uses a case study of a 72-apartment complex to analyse a range of tariff options for distributing the benefits of a shared solar photovoltaic system connected to an embedded network. The analysis of building-wide benefits demonstrates that, for this real apartment complex, the combination of an embedded network and a shared photovoltaic system sized at approximately 1 kW per apartment can generate significant savings to be distributed between apartment owners and residents, in line with previous theoretical analysis [8].

However, the apartment level analysis shows that designing a tariff mechanism to distribute these benefits while satisfying the quadrilemma of cost-recovery, efficiency, consumer acceptance and universal benefit presents a significant challenge. The diversity of household load profiles across the apartment building lowers the building peak demand and therefore contributes to the value generated by the embedded network. However, the important and novel finding of this research is that this diversity also creates challenges for equitable distribution of the benefits and, in particular, prevents the use of cost-reflective pass-through tariffs to deliver universal benefit.

While this tension between the aims of the quadrilemma is common to many tariff design scenarios, the study also highlights a particular constraint placed on embedded network tariff design as an unintended consequence of regulation designed to promote customer choice. The potential for cost-reflective tariffs to result in bill increases for some customers who do not, or cannot, reduce their peak demand, has been noted elsewhere [55]. However, in an embedded network, the right of all customers to access the retail market means that adversely affected customers who cannot reduce their peak demand are likely to leave. Moreover, being constrained to a building or group of buildings, the embedded network has a limited pool of potential customers and cannot survive significant desertion. The requirement for universal benefit therefore acts as a hard constraint on tariff design. Perversely, this effectively prohibits tariff arrangements (particularly application of peak demand charges) that incentivise socially beneficial behaviours within the embedded network, including load shifting to reduce network peak demand.

This finding has implications for the design of tariffs and embedded

network business models and for regulation and policy approaches related to renewable energy communities. There are similarities with the situation in an Integrated Community Energy System (ICES) where community acceptance is required to procure investment in renewable energy infrastructure and is dependent on perceived fairness [56], although the particular differentiation between the roles played by owners / investors and residents / customers is a feature specific to the strata governance arrangements of apartment buildings.

Although based on data from a real apartment complex, a limitation of the study is that data was collected from only 28 of 72 apartments. In particular, some uncertainty is introduced by the assumption that the distribution of demand peaks of the metered apartments is representative of all the apartments. Additionally, in this analysis, household contribution to the building peak has been used as a proxy for cost-reflectivity. Further analysis would benefit from the development of an improved cost-reflectivity metric that includes a consumer's contributions (fixed, volumetric and peak demand) to the bill paid at the parent meter, the lower cost of photovoltaic energy and an additional element for repayment of infrastructure costs. The list of potential tariff arrangements tested in this study is by no means exhaustive and future research might fruitfully explore a wider range of options. Arrangements such as critical peak pricing or peak time rebate might be more effective in incentivising efficient behaviours, although these may not be acceptable to consumers as the former involves high risk and both reduce bill certainty. It would also be of interest to include the impact of household demand response in the modelling, particularly where automation of significant loads such as electric hot water, air conditioning or electric vehicles could significantly increase solar self-consumption, reduce peak demand and improve outcomes for all consumers.

While the case study presented here uses a potential embedded network in an Australian apartment building, the findings also have implications for multi-family housing, energy communities and residential embedded networks or micro-grids in other jurisdictions. Further exploration of the implications under different existing and potential regulatory frameworks would be valuable. Moreover, although the particular findings are specific to residential buildings, the analytical framework could be usefully applied to the design of tariffs and business models for distributing renewable energy in commercial embedded networks.

CRedit authorship contribution statement

Mike B. Roberts: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Arijit Sharma:** Data curation, Formal analysis, Investigation, Visualization. **Iain MacGill:** Supervision, Investigation, Validation, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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